# Baloyi Thabang Bongani Junior

Resume

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## EDUCATION

University of the Witwatersrand — Braamfontein, ZA
Doctor of Philosophy, Mathematical Statistics & Actuarial Sciences (Full Dissertation) — Jan 2027 – Present

- Dissertation: "TBD"

University of the Free State — Bloemfontein, ZA
Master of Science, Mathematical Statistics (Full Dissertation) — Jan 2026 – Oct 2026

- GPA: 80% (WITH DISTINCTION)
- Dissertation: "Understanding Health Insurance Cost Predictions Using Deep Neural Networks, Monte Carlo Simulation and Shapley Values"
- UFS Completion Letter | UFS Academic Record | MSc Dissertation

University of Cape Town — Cape Town, ZA
Bachelor of Science Honours, Statistics and Data Science, minor Actuarial Sciences — Graduated Dec 2025

- GPA: 74%
- Thesis: "Mean-Variance Hedging with Neural Networks using Vector AutoRegression (VAR) Models as the Underlying Sample Path Generator"
- Supervisors: Yuri Robbertze and Jake Stangroom
- Major Courses: Analytics (Supervised & Unsupervised Learning), Multivariate Analysis, Statistical Computing, Actuarial Financial Economics, Portfolio Theory, Operational Research, Time Series, Advanced Probability Theory, Likelihood, Bayesian Analysis
- UCT Transcript | Proof of Qualification | Thesis Repository

University of Cape Town — Cape Town, ZA
Bachelor of Commerce, Management Studies — Graduated Mar 2023

- GPA: 70%
- UCT Transcript | BCom Certificate

University of the Free State — Bloemfontein, ZA
Bachelor of Science, Actuarial Sciences (Incomplete) — 2016 – 2017

- Status: Not completed; first-year coursework passed, several modules with distinction
- UFS Academic Record

## RESEARCH

MSc Dissertation — Jan 2026 – Oct 2026
University of the Free State, Department of Mathematical Statistics & Actuarial Science — Bloemfontein, ZA

- Title: "Understanding Health Insurance Cost Predictions Using Deep Neural Networks, Monte Carlo Simulation and Shapley Values"
- Training a bias-free deep neural network to predict health insurance costs, combining Monte Carlo simulation for input uncertainty quantification and Shapley values for model-agnostic interpretability
- Integrating a three-layer empirical framework (predictive, distributional, attributional) to bridge deep learning prediction with explainable actuarial modelling
- Areas: Health insurance cost prediction, deep neural networks, Monte Carlo simulation, Shapley values, explainable machine learning, actuarial modelling
- Supervisor: Dr JM Blomerus
- Dissertation: Full Text (PDF) | GitHub Repository

BSc Honours Thesis — Feb – Nov 2024
University of Cape Town, Department of Statistical Sciences — Cape Town, ZA

- Title: "Mean-Variance Hedging with Neural Networks using Vector AutoRegression (VAR) Models as the Underlying Sample Path Generator"
- Co-authored with Ditiro Letsoalo; supervised by Yuri Robbertze and Jake Stangroom
- Proposed a novel approach integrating neural networks with VAR models for mean-variance hedging in incomplete markets
- Demonstrated lower hedging errors and better risk-return trade-offs compared to traditional methods through extensive simulations and empirical analyses
- Areas: Mean-variance hedging, neural networks, VAR models, financial risk management, multivariate time series
- Repository: GitHub

## ACADEMIC JOURNALS

- Baloyi, T.B.J. "Understanding Health Insurance Cost Predictions Using Deep Neural Networks, Monte Carlo Simulation and Shapley Values." Manuscript in preparation, based on MSc dissertation, University of the Free State, 2026.

## TEACHING EXPERIENCE

Lecturer (Mathematics) — Aug 2025 – Dec 2025
Cape Peninsula University of Technology, Department of Maritime Studies — Cape Town, ZA

- Took over the Mathematics (Vector Calculus I) module of the Bachelor of Marine Engineering programme mid-year, preparing a large cohort for the final examination
- Identified and closed learning gaps left after the previous lecturer's departure while completing the remaining syllabus under a constrained timeframe
- Prepared lecture notes and tutorials; set the main and reassessment summative examinations; marked tests and examinations and submitted all marks on time
- Delivered structured lecture sessions and provided academic support to students, often beyond standard working hours

Teaching Assistant / Tutor — 2022 – 2024
University of Cape Town, Department of Mathematics and Applied Mathematics — Cape Town, ZA

- Ran weekly tutorial sections for first- and second-year mathematics and engineering mathematics courses
- Invigilated tests and examinations, marked assessments, and provided academic support to undergraduate students across multiple courses

Courses tutored:

- 2024: MAM1004F (Mathematics 1004), MAM1006H, MAM1012F (Mathematics 1012), MAM1021S (Mathematics 1B for Engineers), MAM1031F/33F, MAM2012S (Differential Equations 2DE), MAM2083F (Vector Calculus for Engineers), MAM2084F/S (Linear Algebra & DE for Engineers), MAM2085F (Vector Calculus for Aspect)
- 2022 & 2023: MAM1004F (Introduction to Functions and Calculus), MAM1021S (Mathematics 1B for Engineers)

## TECHNICAL SKILLS

Programming: Python (TensorFlow, PyTorch, NumPy, pandas, scikit-learn, matplotlib), R, SQL, MATLAB, STATA, Java, VBA, LaTeX
Statistical & Mathematical: Statistical Modelling, Machine Learning, Deep Learning, Time Series Analysis, Multivariate Analysis, Bayesian Analysis, Stochastic Calculus, Portfolio Theory, Optimisation
Tools & Platforms: Git, GitHub, Jupyter Notebook, VS Code, AWS, Azure, Docker, Power BI, Tableau, Microsoft Excel

## HONOURS & AWARDS

- Dean's List, University of the Free State
- Best Aggregate Average Actuarial Science, University of the Free State
- Best Aggregate Average Actuarial Science, Moshal Scholarship
- GoldenKey International Society
- Top 3 Learner at Nghonyama High School, Grade 1–12
- Moshal Scholarship
- Jürgen Klauss Bathe Leadership Scholarship
- UCT Postgraduate Funding Scholarship

## ACADEMIC SERVICE & VOLUNTEERING

- UCT Mathematics Competition – Volunteered as organiser/invigilator, 2019, 2023
- Tutor, Grade 8 and 9 – Adopt-a-school Foundation, 2021

## REFERENCES

Ms Samkelisiwe Ntuli ntulisa@cput.ac.za | +27 21 440 5769
Acting Head of Department, Department of Maritime Studies, Cape Peninsula University of Technology

Thomas van Heerden thomas.vanheerden@uct.ac.za | +27 21 650 3200
Senior Lecturer, Department of Mathematics and Applied Mathematics, University of Cape Town

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Departmental Secretary, Department of Maritime Studies, Cape Peninsula University of Technology

Reference letters available upon request.